EduTec

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Public Disclosure Statement

A collection of educational Virtual Reality experiences, aimed at young children aged 11-13, with a focus on experiential learning, equal opportunity, and interactivity.
The product is available on a subscription, with various subscription types based on the customers needs.

The Problem/Opportunity

"If knowledge comes from the impressions made upon us by natural objects, it is impossible to procure knowledge without the use of objects which impress the mind"
– John Dewey (Dewey, 1916)

Traditional education, particularly in middle schools, often relies on students listening to and retaining information, with little explanation as to why it is important, or how it connects to other fields and the world around them. Memorizing and reciting facts and numbers does little to engage the brain, and doesn't allow room for excitement, joy, wonder or creativity. Additionally, lacking awareness of the context that the information lies in reduces information retention and lower understanding.
This method of education is considered to be 'boring' by some students, particularly younger ones, which has a negative effect on overall education.

Unengaged students may not end up learning much at all, and may quickly forget all information at the end of the school year, when the information is no longer required or relevant. Furthermore, completing education without valuable knowledge can cause more issues later on, with students possibly not reaching their full potential.

This issue comes from a general lack of hands-on/experiential learning in the classroom. Little merit is given to the educational benefits of experience and practical skill learning. The knowledge gained in this manner can create long-lasting impacts on a student, as well as being transferable to other subjects.
The benefits of experiential learning is apparent in other areas of education such as tertiary study, degrees may include compulsory practical internships or placements.
For a student to go somewhere, learn something and then apply it in practice is invaluable to their overall education. (Masten, n.d.)

Education also suffers from a lack of technology being efficiently used in the classroom. Technology is a novel way to engage students, providing fun and exciting new ways to learn.
Lack of use is mainly caused due to the high cost of certain technologies, particularly those on the cutting edge such as VR and AR.
This disproportionately affects low-decile schools with lower budgets, who hardly have enough budget to operate as it is, let alone to pay for expensive technologies which are often considered to be "non-essential".
However, the benefits of technology are huge, for one, it can instil a passion and excitement for technology in the user, and can do the same for the content being taught.
Disproportional effects on low decile schools in turn disproportionally affects the local community, of which schools are a key element. This problem now presents itself as not just an education issue, but a socioeconomic one.
In its essence, technology in education suffers from inequal accessibility.

While education suffers from these issues on a global scale, remedy efforts should be made on a local scale, in order to support local curriculums, values and culture. Technology currently used in New Zealand schools are run by overseas companies which do not support the teaching of local knowledge, curriculums and the cultural importance of that knowledge.

This problem became apparent to me through my own personal educational journey. I found myself dreading the classes that had lots of writing, memorization, listening etc. as I found it very unengaging and I often became distracted and bored. Compare this to the anticipation I had for lab classes, and the overall positive effect it had on my understanding of the class content. For example, I had a lot of difficulty in learning about aqueous solutions in chemistry classes, until we did a self-directed in-class experiment, bringing the concepts we had learned about in class to life.
The American philosopher John Dewey had a lot to say about the benefits of experiential learning, stating:

> "On the intellectual side, the separation of "mind" from direct occupation with things throws emphasis on things at the expense of relations or connections" (Dewey, 1916).

These benefits have in themselves been a concept since 350BC. Aristotle once stated:

> "for the things we have to learn before we can do them, we learn by doing them". (Aristotle, 2000)

We have now identified a few problems:

- A lack of experiential learning in education
- Lack of technologies, especially new ones, used in the classroom
- Technologies that are used in the classroom typically have a high cost, and lack localised content
- This high cost creates disproportional usage in favour of high decile schools

Customer & Market

The customer for this product is primarily children aged 10-13. However, a benefit of the product is the capability of ongoing content creation and the wide range of different kinds of content for different age groups and demographics.
Content age recommendations will be defined by the difficulty and maturity of the knowledge being delivered.
The choice of the primary focus on 10-13 year olds is for a few reasons. Firstly, there have been concerns around the health risk of VR on children, especially those under the age of 11. The second reason is that people mature rapidly through these ages. A VR experience designed to be enjoyed by a 11 year old is unlikely to be enjoyed by a 14 year old (again, this can be remedied through additionally produced content for 14 year olds).
Another problem with targeting over 13 year olds is an overall lack of enthusiasm about learning anything, no matter the delivery method.
There is also the problem of aligning with school curriculums, and ensuring the content is of the correct maturity and difficulty.
During development, contact can be made at local schools with students of this demographic in order to test out content as well as technology, and health effects of short term VR use.
The market itself that I intend on targeting is a mix of schools, and personal at-home use, primarily focusing on the former. This allows for education to be brought into the home, where children can learn as well as being a fun activity to do at home.
Another potential market is the homeschooling market. Although it wouldn't be too different from the casual home use market, its worth acknowledging homeschooled students and marketing directly to them. There are currently 6008 homeschooled students in New Zealand, and to create a wide awareness of the product within that market would be financially beneficial.
The schools market is massive, with 2,534 schools in Aotearoa New Zealand alone, (*Ministry of Education - Education Counts*, n.d.) with a huge potential for growth as awareness of the product spreads between schools, and the option of moving internationally, even somewhere as close by as Australia, adds 9,389 extra schools into the mix. (*National Education Report*, 2015).
The market continues to grow as long as there are more schools to market the product to. Personal use is also a large market, with educational apps being the second most popular category on the Google Play store. (*Google Play Most Popular App Categories 2020*, n.d.)
This shows that there is high demand for educational apps, but there is also a high amount of competition, however this product stands out due to the VR element, and will be heavily promoted through its use in school classrooms.
The potential for growth in this market is also huge, and will grow alongside its classroom usage.
These insights were gained through my work experience at a company who work in children's education.

Alternatives

The use of new technologies in the classroom is not a new concept, and the use of Virtual Reality and Augmented Reality is no exception.

One of the largest products in the Virtual Reality classroom space is ClassVR, a USA based company which creates Augmented Reality, Virtual Reality and Mixed Reality content to be used in the classroom. The content is aligned with US State standards and works with local curriculums, and boasts over 500 pre-made activities.
ClassVR does very well in the United States and is available in other countries, including Australia. The headsets are available in New Zealand but it is unknown whether they are in use in classrooms. Their content is designed to be delivered alongside USA curriculums, which would make it inappropriate for New Zealand students.
For example, a United States history lesson may cover subjects such as the Civil War, but wouldn't touch important New Zealand history such as the signing of Te Tiriti o Waitangi. There is also value in students seeing the country they inhabit represented in the VR lesson, for example learning about New Zealand flora and fauna, and being able to go outside and apply that knowledge. There is also value in using and teaching both English and Te Reo Māori words, for instance, during a class about New Zealand birds, the English word for each word could be taught alongside the Te Reo Māori word, e.g. Morepork, and Ruru.
ClassVR also uses prebuilt standalone VR headsets, which adds extra cost to the customer, especially if purchasing for an entire classroom of students.
A set of 8 headsets costs $5600 NZD on pbtech.co.nz.

Another excellent example of VR for the classroom is Google Expeditions, which has been designed for the Google Cardboard viewer. Google Expeditions focuses on immersion, taking the user all over the world through 360° images, and being able to click on various objects on that image to learn more about it.
There are over 900 VR Expeditions available, such as a trip to Mt Everest, or back in time to the Battle of Hastings.
Google Expeditions is free to download, and a Cardboard viewer can be purchased for as little as $5, or even constructed at home with basic resources. The only hardware required is a smartphone, and the app runs even on low specification phones.
However, as low-cost Google Expeditions is, it comes with the drawback of once again simply feeding information to the student without any application of that knowledge. Although credit can be given for making content delivery more novel, exciting and immersive, it still suffers from the same issues in education as previously outlined.

Solution

The product that I have to offer in order to take the opportunity created by the previously outlined problem is as follows:
A collection of educational Virtual Reality experiences, aimed at young children aged 11-13, designed to work with local curriculums. There are 4 key elements to this product.

The first of these is Experiential Learning. Experiential learning refers to the process of learning through experience. Virtual Reality gives the capability of placing the student in situations that couldn't be accessed beforehand due to lack of general accessibility and monetary cost. For example, a class of students could experience viewing a rocket launch, without even leaving the classroom. While viewing the rocket launch, there is opportunity to learn about the branches of STEM, as well as providing excitement and immersion.

The second is a focus on interactivity and application of knowledge. The purpose of this is to break free of the common flaws of academic education, to memorise and recite facts and figures. The focus on this allows for a student to not only learn something, but to also apply it, see it in action, and ultimately gain a much more in-depth understanding, with longer-term retainment. An example of this, using our rocket launch example, is the student learning about forces such as thrust and gravity, and using that knowledge to balance the forces on the rocket so that it will lift-off into space.

Thirdly, is a focus on contextually aware content. This involves showing how the content that it delivers relates to other knowledge under different subjects. Doing this strengthens understanding of the content, and how it applies to the world. Students don't just learn *how* something works, but also *why*, and what else it might affect. Once again using the rocket analogy, students may learn about the science behind the combustion of the rockets engine, and also the mathematics of the forces acting upon the rocket as it flies. You could even take it a step further, and teach environmental science through the consideration of the environmental impact that such an engine would create. It is important for students to understand that knowledge is related to something else.

The fourth key element is the products ability to be tailored to the space it operates in. Initially, it will be built for New Zealand audiences, such as learning NZ history, about NZ industry or NZ nature. However, if the product is taken overseas, to Australia for instance, content will have to be created to cover Australian history, industry and nature.

Finally, we aim to ensure that the product is as low-cost as possible, and having a social focus. This is to make sure that the product is not only available to schools and individuals of higher income.
The product is low-cost by nature. Delivery only requires purchasing of the content itself through a subscription model. The target platform is Google Cardboard, meaning that

the only required hardware is a low-specification smartphone. Cardboard viewers can be purchased for as little as $5, and can also be made at home or in the classroom as an extra activity. Development cost is also low, with the low processing power of a smartphone being a barrier against high definition graphics (this is not a barrier against the product itself. Knowledge can be delivered just as effectively with low-polygon graphics, as long as the experience is designed well).
The social focus can be achieved through a 'buy one, give one' model, most notably achieved by the New Zealand company Eat My Lunch, who give a free lunch to low decile schools for every lunch bought from them.
Following this model means that for every subscription purchased by an individual or high decile school would provide the same subscription to a low decile school. Technology, and new, novel ways of learning should not be reserved for those of high income, it should be for everyone of all communities.

Competitors in this market do exist, and do similar work, however they are not locally based and lack the social focus offered by our product. It's a niche, wicked problem with many different possible solutions, which is why this problem remains unsolved.

At this point in time, the product exists as a demo level, produced for a university course. The level takes the user through all 4 branches of STEM, Science, Technology, Education and Mathematics, in the context of launching a rocket into space.
The user completes 4 activities, one for each branch, and finishes the level riding the rocket as it lifts off.
This demo level has been tested and reviewed by Dr Stefan Marks, a senior lecturer at AUT with an area of expertise in Virtual Reality. The feedback provided can be incorporated into future versions of the level.
There is also potential for communication with Ormiston Junior College in Auckland, who are "looking for projects related to STEAM Education for its students featuring year levels 7 – 10 (11 – 15-year olds)". Collaboration with Ormiston allows for user testing, as well as input from teachers in order to make the level as effective at delivering information as possible.
Communication has begun with Ormiston with an opportunity to begin user testing being planned out, but this planning has been interrupted by the uncertainty raised by the resurgence of COVID-19 in the New Zealand community.

In terms of legal setup, Unity game engine pro licenses will need to be purchased, as well as permission to distribute on Google Play and the App Store.
Approval will be sought from the Ministry of Education to operate in schools and to ensure content is in line with the school curriculums.

Advantage

We will be the first New Zealand owned, based and staffed companies in this market. This is considered an advantage due to market knowledge as well as our staff having a better relationship and knowledge with the unique local content we aim to deliver through our product.
While other companies do similar work in New Zealand, they are mostly international companies who have expanded into the market, and a lot of the generated revenue ends up going overseas. This will be a locally produced product, and doing business with us will help to contribute to the local economy. This is becoming ever more important in a post-COVID-19 world, with an increased demand for locally produced content and produce.
The intellectual property belongs entirely to myself. Patents and copyright can also be obtained.

Sales, Marketing & Distribution

The sales strategy for this product is to have all content available for purchase through a subscription model, accessible through a free-to-download app available on Google Play and the App Store.
There are two kinds of subscription: personal and institute. Personal subscriptions can be logged in to on one device at a time, while institute subscriptions allow for multiple logins up to a certain limit on multiple devices. Personal subscriptions cost much less than an institute subscription, but institute subscriptions cost less than purchasing a personal subscription for each individual student, therefore making it more cost-effective to purchase an institute subscription for a classroom.
Different levels of subscription allow for access to different content. For example, a Basic subscription allows access to most content, but a premium subscription will allow access to all content.
The higher the level of subscription, the higher the cost.
Between personal/institute and basic/premium subscription types, there are 4 different combinations of subscription. They are, in order of price: Personal Basic, Personal Premium, Institute Basic and Institute Premium.
There is also a possibility of purchasing different 'packs' for different subjects – a science pack, history pack for instance or even more specific: the science pack becomes a physics pack, which becomes an electronics pack etc.

The social aspect of the product is also an important factor to consider. As previously mentioned, we aim to achieve a 'buy one, give one' model, where every subscription bought by a high-decile school 'gifts' an equivalent subscription to a low decile school, ensuring the product is available to schools of all deciles. This could mean a small drop in income, but seeing as low-decile schools are unlikely to purchase a subscription if its out of budget, the drop is negligible.
The goal of this product is not to generate income, it is to keep financially stable while bringing technology into the classroom, improving the state of education and ensuring accessibility for all.
We aim to market the product mainly directly to schools, as they are the primary target audience. Personal subscriptions will be marketed through the schools as well. If the product is popular with the individual student, they will be more inclined to ask for a personal subscription to be used at home.
Additionally, use can be made of Googles AdSense program to increase product awareness. This will be used to promote personal subscriptions, as well as being the only way of marketing towards the homeschooled student demographic.

Finance

The key stream of revenue for the product is the ongoing subscription fee, either charged monthly or annually. Most of that revenue comes from schools. Additional smaller revenue can be obtained from personal subscriptions.

Startup costs can be obtained through grants. Virtual Reality and other 'future technologies' are largely popular currently and there are lots of grants on offer around that industry. When we combine this with our social and local focus, we have two highly marketable aspects to our product which can be used to secure grants and funding. We can also approach the Ministry of Education, and attempt to gain funding from them. There are also opportunities to gain funding from Callaghan Innovation, and the Ministry of Business, Innovation and Employment.

According to the spreadsheets, we will begin operating at an overall profit after about 10 months. If we continue making that profit we will be out of debt after 21 months, assuming we operate in 60 schools countrywide. Once we begin to make a profit, expansion is key.
Operating in as many schools as we can will bring in a lot of revenue, which can be used to create more content, polish existing content, and move into other markets such as tertiary education, other countries and even, long-term, new technologies and products.

The main risks to our success is not generating enough interest about the product when its in development. The first six months are dedicated to development and marketing, and will not be creating any income until afterwards. If insufficient sales occur after launch, we will be unable to repay any loans and will fold. Therefore, marketing is key to financial success.

The Team

The team currently only consists of myself.

My name is Liam Coll, and I am a third year Bachelor of Creative Technologies student with a minor in game design.
I am on track to graduate at the end of 2020.
I am multi-skilled, but my area of expertise is in game development, and I have solid experience using the Unity Engine, which I used to produce a demo level for this product. I also have skills in innovation and entrepreneurship, as well as UX design.

In order to get this product off the ground, I will need a small team of hard-working passionate individuals. A realistic small team would consist of:

- 3 developers to create and test content.
- 2 environment/3D artists to assist developers with producing the necessary assets.
- 2 content designers, to design content to be in line with local curriculum, and that knowledge is efficiently communicated. Ideally would have experience teaching or working in education previously.
- 1 CEO/project manager, to take care of finances and ensure the development runs smoothly.
- 1 marketer, to create product awareness and to be in charge of sales and distribution.

Milestones

We have produced a proof-of-concept demonstration level, which aims to teach the definition of STEM through a rocket launch. This level has been play-tested and reviewed by a senior lecturer at AUT. Having this level is great advantage when asking for funding, as we have a tangible output that we can point at, and allows us to fully communicate what the product is, and why it's important and a worthwhile investment.

If we were to win a prize in X Challenge, we would set the prize money aside in a separate fund and invest while we go about securing extra funding, investors and grants in order to make the company/product a reality.

Appendix

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Eat My Lunch

ClassVR

Link to demo level

Link to footage of demo level on Android